Q.1 (2 Marks Each)

1. Define the following terms: Bug, Defect, Error, Failure, and Fault.
2. List and briefly define any four Black Box Testing techniques.
3. Explain the need for software metrics in the testing process.
4. Differentiate between Static Testing and Dynamic Testing.
5. List the key components of a test plan as per the IEEE 829 standard.
6. List the attributes and the corresponding metrics used for monitoring and controlling the testing process.
7. Differentiate between Verification and Validation.
8. What is cyclomatic complexity, and provide its formula.
9. Define Alpha Testing and Beta Testing.
10. Explain the terms Stub and Driver in the context of Integration Testing.
11. State the goals of software testing.

Q.2 (5 Marks Each)

12. Illustrate the Software Testing Life Cycle with the help of a neat diagram, and explain each phase with a suitable example.
13. Apply Boundary Value Analysis and Equivalence Class Testing to design test cases for a program that accepts three sides of a triangle, each ranging from `1 to 100`, and identifies whether the triangle is Scalene, Isosceles, Equilateral, or Invalid.
14. Analyze the different Regression Testing techniques and evaluate their applicability with a suitable example scenario.
15. Evaluate different Test Suite Prioritization techniques for a given project. Compare their effectiveness and discuss the metrics used to measure prioritization effectiveness.
16. Explain the Software Defect Life Cycle with a neat diagram and briefly define all its states.
17. Describe the four Integration Testing strategies with diagrams, and explain the role of Stubs and Drivers in each.
18. Explain Basis Path Testing. For a given control flow graph, compute the cyclomatic complexity and determine the number of independent paths.
19. Analyze the Test Planning process in detail. Explain how Test Organization, Structure of the Testing Group, Detailed Test Design, and Test Specification contribute to effective test management. Illustrate your answer with a sample test plan outline.